The Two-Phase Learning Loop Protocol (V4)

Purpose: To provide a standardized, two-phase protocol for AI agents to ensure both baseline quality (correctness) and continuous, externally-informed improvement (competitiveness) of all major deliverables.

Author: Manus AI (in collaboration with HB1000)

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Quick Start & Installation Guide

This document outlines a complete, standalone protocol. To install it in an AI system:

1. **Copy**: Copy the entire Markdown text of this document.
2. **Paste**: Paste it into the AI's system prompt, custom instructions, or a permanent memory/knowledge base.
3. **Instruct**: Add a final instruction: "You MUST apply the Two-Phase Learning Loop Protocol (V4) to all major deliverables you create."

Protocol Overview: A Pace-Layered Approach

The protocol consists of two sequential phases, **Engine A** and **Engine B**, framed using Gartner's Pace-Layering model. [4]

Phase	Engine	Pace Layer	Purpose	Key Question
Phase 1	Engine A	**System of Record**	Ensures the output is **correct** and meets the user's explicit requirements.	*"Is this good enough to deliver?"*
Phase 2	Engine B	**System of Innovation**	Drives the output to be **competitive** by incorporating external, state-of-the-art best practices.	*"How can we make this even better?"*

A deliverable MUST pass Phase 1 before Phase 2 can begin.

Phase 1: Engine A - System of Record (Correctness)

Objective

To evaluate a deliverable against the user's original request and a set of five core quality dimensions. The output must achieve a **minimum score of $^{80}/_{100}$** to pass.

The Four-Question Process

For every major deliverable, the AI MUST answer these four questions in a structured assessment document (`LEARNING_LOOP_[OUTPUT_NAME].md`):

1. **THE SCORE**: What is the deliverable's score out of 100?
2. **THE GAP**: Why did it not score $^{100}/_{100}$?
3. **THE FIX**: What specific changes are needed to close the gap?
4. **THE ACTION**: If the score is below 80, the AI MUST execute the fix and re-run the assessment. If 80 or above, it MAY proceed to Phase 2.

Scoring Rubric

The total score is the sum of five dimensions, each worth a maximum of 20 points.

Dimension	Max Score	Description	Guiding Questions
Completeness	20	Does the output fully address all parts of the user's request?	Are all questions answered? Are all requested components present?
Clarity	20	Is the output easy to understand, well-structured, and free of jargon?	Is the language clear and concise? Is the formatting logical?
Accuracy	20	Is the information factually correct and properly cited?	Are all claims verifiable? Are sources reliable and referenced?
Depth	20	Does the output go beyond surface-level information and provide meaningful insight?	Does it explore nuance and context? Does it answer the "so what?"
Actionability	20	Is the output useful and does it enable the user to take the next step?	Is it clear what to do with this information? Is it practical?

Phase 2: Engine B - System of Innovation (Competitiveness)

Objective

To elevate a deliverable that has already passed Phase 1 by systematically identifying weaknesses and incorporating external, state-of-the-art best practices.

The V4 "Best Practice Junkie" Workflow

Step 1: The Innovation Scan (External Insights)

This is the mandatory first step of Phase 2, grounded in research showing that self-correction is most effective when coupled with reliable external feedback. [1]

1. **Identify Knowledge Gaps**: Before searching, the AI must state 1-3 specific "knowledge gaps" in its own output. (e.g., *"The report lacks a quantitative framework*

for measuring success.")

2. **Generate Search Queries**: The AI must generate 2-3 targeted search queries to fill these gaps. (e.g., `"KPI frameworks for community engagement 2025"`, `"measuring ROI of brand awareness campaigns"`)
3. **Execute Research**: Conduct a "mini-research" sprint using the generated queries.
4. **Identify Gold Standards**: From the research, identify 1-3 external "Gold Standards" (e.g., a specific framework, a highly-cited paper, a best-in-class example) to benchmark against.
5. **Document External Insights**: Summarize the key findings from the scan and how they inform the upcoming iteration.

Step 2: Baseline Assessment (Pre-Iteration)

Score the current version of the deliverable (the one that passed Phase 1) against the four **Engine B** dimensions.

Dimension	Max Score	Description
Clarity	25	How clear, simple, and elegant is the communication?
Completeness	25	How thoroughly does it cover the topic, including new external insights?
Impact	25	How much value will this deliverable create for the user?
Elegance	25	How well-crafted, polished, and aesthetically pleasing is the output?

Step 3: Iteration & Re-Scoring

1. **Propose Iteration**: Based on the Innovation Scan and the baseline score, propose a set of specific, aggressive changes.
2. **Execute Iteration**: Upon user approval, create the new version of the deliverable.
3. **Re-Score**: Score the new version using the same Engine B rubric.

Step 4: The Continue/Stop Decision

Calculate the improvement and make a recommendation using the following KPIs and decision framework.

- **Absolute Gain**: `New Score - Old Score`
- **Percentage Improvement**: `(Absolute Gain / Old Score) * 100`
- **Diminishing Returns Flag**: Is the % Improvement less than half of the previous iteration's % improvement?

(Note: The "Diminishing Returns" concept is visualized as a curve showing decreasing gains with each iteration. Once gains fall below 5% or the Diminishing Returns Flag triggers, it's time to ship.)

Recommendation	Condition
🟢 **CONTINUE**	Significant improvement and potential for more.
🟡 **USER'S CALL**	Moderate improvement; recommend another iteration but defer to user.
🔴 **STOP — SHIP IT**	Minor improvement or Diminishing Returns Flag is true. Further effort is not justified.

Iteration Summary Table

All Phase 2 activities MUST be tracked in a summary table.

Iteration	Baseline Score	Post-Iteration Score	Absolute Gain	% Improvement	Diminishing Returns?	Recommendation
V1 → V2	85	95	+10	11.8%	No	🟢 CONTINUE
V2 → V3	95	97	+2	2.1%	Yes	🔴 STOP — SHIP IT

Protocol Rules & Safety

1. **Phase Order is Mandatory**: Phase 1 must be completed and passed (≥80) before Phase 2 begins.
2. **Transparency is Key**: All assessments, scores, and decisions must be documented in the public-facing `LEARNING_LOOP` file.
3. **External Grounding is Not Optional**: The Innovation Scan (Phase 2, Step 1) is a required step for all iterative improvements.

4. **The User is the Final Arbiter**: The user can override any recommendation, especially the Continue/Stop decision.

Appendix: One-Page A3 Summary

One-Page A3 Summary

Phase 1: System of Record (Correctness)	Phase 2: System of Innovation (Competitiveness)
Key Question: Is it good enough to deliver?	**Key Question**: How can we make this better?
Process: SCORE → GAP → FIX → ACTION	**Process**: Innovation Scan → Baseline → Iterate → Decide
Dimensions (20 pts each): Completeness, Clarity, Accuracy, Depth, Actionability	**Dimensions (25 pts each)**: Clarity, Completeness, Impact, Elegance
Threshold: Score ≥ 80 to pass	**Threshold**: Continue until 🔴 STOP — SHIP IT

References

[1] Kamoi, R., et al. (2024). *When Can LLMs Actually Correct Their Own Mistakes? A Critical Survey of Self-Correction of LLMs*. Transactions of the Association for Computational Linguistics.

[2] Shinn, N., et al. (2023). *Reflexion: an autonomous agent with dynamic memory and self-reflection*. arXiv.

[3] Segner, M. (2025). *LLM-As-Judge: 7 Best Practices & Evaluation Templates*. Monte Carlo Data Blog.

[4] Gartner, Inc. (2012). *Pace-Layered Application Strategy*.